

		height from surface, $h = 4.231 \times 10^7 - R$
11	A	$M d \omega^2 = \frac{GM(3M)}{(d+D)^2} \dots\dots(1)$ $3MD \omega^2 = \frac{GM(3M)}{(d+D)^2} \dots\dots(2)$ Compare (1) & (2), $d = 3D \dots\dots(3)$ $\text{From (2)} \quad \omega^2 = \frac{GM(3M)}{(3D+D)^2} \left(\frac{1}{3MD} \right)$ $\omega^2 = \frac{GM}{16D^3} \dots\dots(4)$

		$ \begin{aligned} KE \text{ of 2 stars} &= \frac{1}{2} M (d \omega)^2 + \frac{1}{2} 3M (D \omega)^2 \\ &= \frac{1}{2} M (3D)^2 \omega^2 + \frac{1}{2} 3M (D \omega)^2 \\ &= \frac{9}{2} M D^2 \omega^2 + \frac{3}{2} M D^2 \omega^2 \\ &= 6 M D^2 \omega^2 \\ &= 6 M D^2 \left(\frac{GM}{16D^3} \right) \\ &= \frac{3GM^2}{8D} \end{aligned} $
12	D	Wrong answers: $ \begin{aligned} \text{(B) Total } KE &= \frac{GM(3M)}{2(3D)} = \frac{GM^2}{2D} \\ \text{(C) Total } KE &= \frac{GM(3M)}{2D} = \frac{3GM^2}{2D} \\ \text{(D) Total } KE &= \frac{GM(3M)}{2(3D)} + \frac{GM(3M)}{2D} = \frac{2GM^2}{D} \end{aligned} $ Same temperature implies thermal equilibrium. Hence all 3 objects are in thermal